



Department of Electrical Engineering

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High Electricity Bills Due to Inefficient Household Appliances

Group 12

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Project Background and Scope

Our project focuses on developing a **Smart Energy Monitoring System** to address high electricity bills caused by inefficient household appliances. The system enables real-time appliance-level monitoring, accurate energy calculation, and alert notifications. Our team contributed by modeling the system in **MATLAB/Simulink**, designing the **PCB**, and creating the enclosure using **SolidWorks**.

Work Plan & Timeline

1. Problem Identification & Literature Review

The project begins by identifying the problem of high household electricity consumption and reviewing existing smart energy monitoring systems, IoT technologies, and related research.

2. System Architecture & Design

Monitoring, power and energy calculation, alert mechanism, and IoT communication framework.

3. MATLAB/Simulink Modeling

Household appliances and system components are modeled in MATLAB/Simulink to represent real-time power consumption behavior

4. Power, Energy & Alert Logic Implementation

Instantaneous power, energy accumulation, electricity cost calculation, and threshold-based alert mechanisms are implemented and integrated.

5. Simulation Testing & Analysis

The complete system is tested under different operating conditions to validate accuracy, reliability, and alert functionality.

6. Hardware, PCB & Enclosure Design (SolidWorks)

Hardware schematics, PCB layout, and mechanical enclosure are designed, including protection circuits and thermal considerations.

7. Report Writing & Final Submission

Simulation results, design methodology, and conclusions are documented, and the final project report is prepared for submission.

System Development & Implementation

1. Design Philosophy

The design focuses on appliance-level monitoring to identify devices that waste energy, unlike traditional meters that show only total consumption. A modular and scalable design is used, allowing new appliances to be added easily. By using IoT technology, the system provides real-time information and instant feedback to users.

2. System Architecture & Modeling

The system was first modeled and tested using **MATLAB/Simulink** before hardware construction. It consists of four main functional blocks:

- **Monitoring Units:** Represent individual household appliances and their power ratings.
- **Calculation Module:** Calculates instantaneous power, total energy consumption, and electricity cost.
- **Threshold & Alerts:** Continuously monitors energy usage and activates alerts when preset safety limits are exceeded.
- **IoT Layer:** Simulates MQTT data transmission to a cloud dashboard for remote monitoring and visualization

3. Hardware Implementation

The physical system combines custom electronic circuits inside a protective enclosure.

- **Circuit Design:** An ESP32 microcontroller is used to control the relays. Flyback diodes (1N4007) and transistors are included to protect the controller from high-voltage spikes.
- **PCB Layout:** A custom two-layer PCB contains the AC–DC power supply and relays. Proper isolation distances are maintained between high-voltage AC mains and low-voltage DC circuits to ensure safety.
- **Enclosure:** An ABS enclosure designed using SolidWorks houses all components. Thermal simulations showed that natural air cooling is sufficient to maintain safe internal temperatures (around 28 °C) without the use of cooling fans.

Challenges & Resolutions

1. Protecting the Chip from Voltage Spikes

The system uses relays to switch appliances on and off. However, relays create high-voltage "spikes" (back-EMF) when they switch, which can instantly destroy the delicate ESP32 microcontroller.

- **Solution:** design a protection circuit using flyback diodes to block these spikes.

2.Overheating in a Sealed Box

The ESP32 chip gets hot when using WiFi, and the power supply also generates heat. Since the enclosure is sealed plastic with no fan, the heat can get trapped.

- **Challenge:** You have to prove the heat will escape naturally so the box doesn't melt or damage the electronics
- **Solution:** The internal layout allows natural airflow. Hot parts like the ESP32 are placed at the bottom, so warm air rises and cool air moves down along the walls. The PCB is mounted on standoffs to let air flow underneath.

3.Safety with High Voltage (PCB Design)

The PCB was specially designed to safely place both the high-voltage AC power supply and the low-voltage DC control chips on the same board.

- **Solution:** A custom two-layer PCB was designed with strict **isolation gaps**.

4. Simulating Real Internet Connections

Connecting real sensors to the cloud is complicated because of unstable WiFi or network delays.

- **Solution:** Instead of struggling with a real, unstable WiFi connection during the initial design, the team built a **simulated IoT layer** using a Zero-Order Hold (ZOH) block in MATLAB

Individual Contribution & Team Work

The work presented in this project was carried out through a collaborative effort among all group members, with clearly defined responsibilities to ensure effective development and documentation of the system.

P.H.N.D.Jayawardana (230296P) was responsible for conducting **the literature review, organizing the technical content**, and preparing the research paper using LaTeX in accordance with the IEEE conference formatting guidelines.

P.G.I.Jithmal (230306B) focused on the **mechanical and hardware prototyping** aspects of the project, designing the enclosure and system layout using **SolidWorks** to represent the physical realization of the proposed system.

T.M.Kannangara (230314X) was primarily responsible for the design, implementation, and **simulation** of the Smart Energy Monitoring System using **MATLAB/Simulink**, including appliance-level modeling, power and energy calculations, threshold detection, and IoT communication logic.

We held bi-weekly meetings to track progress, solve issues, and make sure all modules worked together correctly.

Conclusion & Reflection

Technical Achievements

- Designed and simulated an IoT-based smart energy monitoring system.
- Implemented real-time power, energy, and cost calculation.
- Developed threshold-based alerts and MQTT data communication.
- Designed PCB, enclosure, and validated thermal safety.

Learning Outcomes

- Gained understanding of smart energy monitoring and IoT systems.
- Improved skills in MATLAB/Simulink and embedded system design.
- Learned PCB design, hardware protection, and thermal analysis.
- Enhanced teamwork and technical documentation skills.

Conclusion

This project successfully developed an IoT-based smart energy monitoring system to help reduce electricity bills through real-time, appliance-level monitoring. MATLAB/Simulink simulations confirmed accurate power measurement, energy calculation, cost estimation, and MQTT communication. The hardware design, including a custom PCB and a 3D-modeled enclosure, shows that the system is practical, safe, and suitable for household use.

Reflection

Viability: Thermal simulations showed that the device can operate continuously without overheating. The internal temperature remains safe (around 28 °C) using only natural air circulation.

Design Choice: The results confirmed that a fully sealed enclosure is sufficient, and no cooling fans are needed for safe operation.

Collaboration: The project was successfully completed through good teamwork, combining skills in software simulation, mechanical design, and technical documentation.